Software Requirements and Design Document

For

CodeWorlds

Version 1.0

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Overview (5 points)

Code Worlds is going to be an educational 2d RPG game, inspired by early Pokémon and online education games. Players will control a player character to explore the world, complete quests, including a main storyline, capture CodeMals, and engage in CodeMal battles. Throughout the game, players will have to complete coding minigames to move forwards. The defining feature of Code Worlds will be the coding minigame which may be involved at various points within the game, sometimes within battles or captures, and sometimes during exploration.

Functional Requirements (10 points)

1) Player movement and environment – High priority. The player will control a character and move them throughout the world. This feature is simple and easy, but that does not take away from its importance. Making sure the movement is seamless is paramount.

2) Unique expansive world – Medium priority. The game world shall be engaging to explore and play in, with unique assets, distinct areas, and varied activities distributed throughout a large map

3) Quests and NPC's - High priority. The game shall include one main questline that ends the game when completed, requiring the player to visit every region of the world, complete multiple combat encounters, and work through coding challenges designed to teach programming concepts. The game shall also include side quests that are shorter than the main questline, each containing some combination of combat encounters, exploration objectives, and NPC interactions, to provide variety in activities. NPCs shall be used to give, advance, and complete quests, and shall have dialogue, change location to match quest progress, and give rewards based on the player's progress. Detailed NPC characterization, such as backstory and personality, shall be a lower priority than the main questline and side quests

4) CodeMals – High priority. The mechanics of CodeMals is the beating heart of the game. The battles, capture sequences, and interaction with the coding minigames will be what draws players the most and gives CodeWorlds a unique identityPlayers will collect, battle, however, it remains a sophisticated and difficult challenge because of these things. Users should be able to grow throughout the game with their CodeMals. The mechanics of these systems will be fleshed out over time as we determine what is most fun to interact with, but for now battles and captures will mostly resemble those from Pokémon, with coding challenges for an interesting twist.

5) Coding minigames – High priority. Players will be required to complete coding questions throughout the game in order to progress and succeed. Coding questions connected to questlines may be custom made and more difficult, while those encountered randomly may be procedurally generated through a custom system.

6) UI – Low priority. A basic UI has to exist for basic functionality, but a well designed and good looking UI is a low priority. Things such as a main menu, pause menu, clear buttons and interactables within the world, and other necessary elements will be implemented early on, and polished in the late stages of the project.

7) CodeMal Design – Low priority. Custom assets and designs of the CodeMals is considered a polish stage priority and not essential to the gameplay. Placeholder assets will be used for a majority of development, while custom art will be created later on.

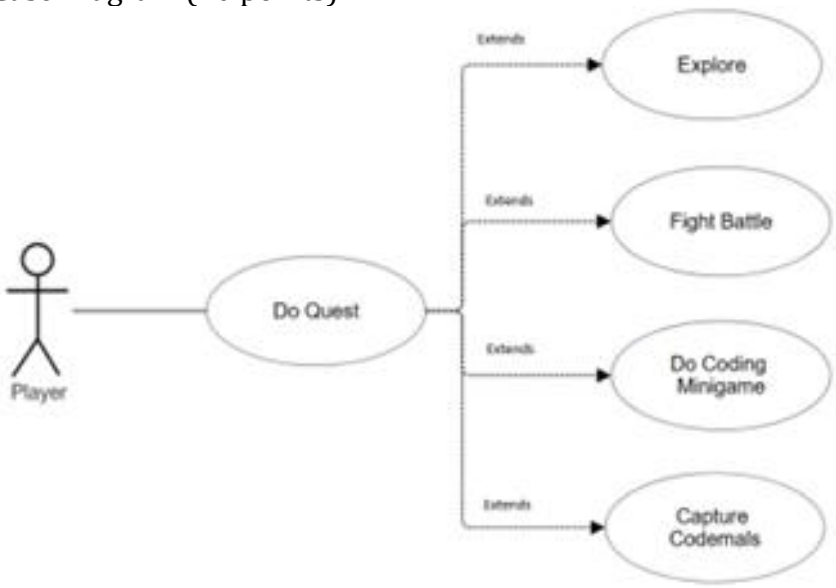
8) Save Game Functionality – Players will be able to save where they are in the game to their local machine, and have multiple saves at the same time to go back and forth between.

Non-functional Requirements (10 points)

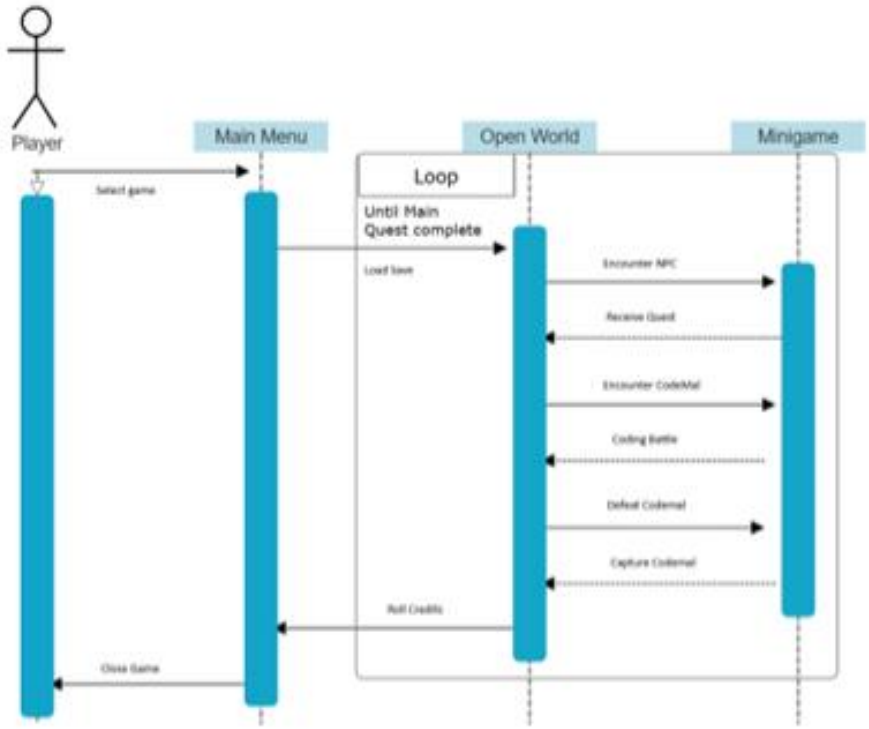
1. Easy to run – This is meant to be a game for anyone to learn how to code and should therefore run on any computer that anyone might use. The graphics will be basic, physics will be mostly nonexistent, and any feature that would jeopardize the ability for someone to learn to code will not be implemented.

2. Entirely local – There will be no need to connect to the internet beyond the download of the game. This will function as a classic single player game, so that someone may learn to code anywhere.

Use Case Diagram (10 points)



Class Diagram and/or Sequence Diagrams (15 points)



Operating Environment (5 points)

This will be a desktop game. Godot supports mac, windows, and linux, so it is intended to work on all 3, though only on modern version of said systems. The game will run locally and save locally, it will not interact with any external systems other than the operating system.

Assumptions and Dependencies (5 points)

This project assumes continued support by Godot, along with the ability for Godot to export games for the 3 listed operating systems. We also assume the existence of a free asset library for development, though we don't plan to keep these assets for the entirety of the project. We also assume that Nintendo doesn't sue us like they do everyone else.